

Left Join vs Right Join

Left Join returns all rows from the left table and matching rows from the right table (non-matching right rows show NULL), while Right Join returns all rows from the right table and matching rows from the left table (non-matching left rows show NULL). They are functionally opposites and can be interchanged by reversing table order.

On clause vs Using clause

ON clause specifies an arbitrary join condition with column names that may differ between tables (e.g., ON t1.id = t2.ref_id). USING clause is used when join columns have the same name in both tables and automatically equi-joins on that column, also removing the duplicate column from the result set.

Natural Join vs Inner Join

Natural Join automatically joins tables based on all columns with identical names, without explicit join condition, and removes duplicate columns. Inner Join requires an explicit join condition (ON or USING) and retains all columns from both tables, including duplicates.

Inner Join vs Left Join

Inner Join returns only rows with matching values in both tables, excluding non-matching rows from either side. Left Join returns all rows from the left table, with matching rows from the right table or NULLs where no match exists.

